

Recursive Coordination Information Theory:

Decision Sufficiency, Consequence Sufficiency, and Recursive Closure

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Abstract

An information interface can preserve every relative action cost required by a current decision and still omit distinctions needed to evaluate the circumstances that action creates. This paper develops Recursive Coordination Information Theory (RCIT) around that separation, within a common Coordination Information Theory (CIT) framework for task-relative information contracts and source contraction. Declared current queries induce a decision interface G_D , while action-conditioned successor queries induce an evaluation interface G_E^+ . Decision information supports the latter requirement exactly when G_E^+ is a function of G_D . For finite deterministic interfaces under a specified source law, $H(G_E^+ | G_D)$ is the minimum conditional information in a supplement that permits evaluation. A same-task construction makes this gap positive despite complete current-regret sufficiency. Classical controlled-Markov refinement supplies recursive completion, whose conditional entropy measures the information added beyond the starting contract. Exact examples distinguish current sufficiency, recursive closure and reward-preserving abstraction, and show that individually closed contracts can require further information when joined. Potential-based reward shaping and finite rational inattention provide connections to invariant task descriptions and the Nash-welfare form of the fixed-slope information-regret problem. Separate recursive-learning investigations examine evaluation provenance, persistence and externally supplied training targets. The resulting framework characterises what an interface must retain for current use, successor evaluation and recursive coordination; learning and acquiring that information remain distinct implementation questions.

Keywords: task-relative information; sufficient representation; coordination; state abstraction; recursive closure; rate-distortion.

1 Information needed now and after action

Consider a network controller comparing two ways of reallocating traffic. Suppose two network states give the same current relative cost for every available action. An interface designed to preserve that comparison can treat the states as identical. Yet implementing the same preferred reallocation may leave different circumstances for the next decision. In one state, a later action remains attractive. In the other, the ordering of the next alternatives reverses. Evaluating that difference requires information that the current comparison could discard.

A finite example makes the issue precise. Let the two current action-regret rows be $(0, 1)$ and $(0, 1)$, with the first action optimal in both. If that action produces successor rows $(0, 1)$ and $(1, 0)$, the complete current regret row does not identify the next one. The current comparison is exact; the missing distinction concerns its action-conditioned successor under the same state-dependent task.

The principle examined in this paper is:

The information sufficient to choose a means need not be the information sufficient to evaluate the circumstances created by using it.

Choice can already involve extensive prediction and consequence evaluation. The question is whether the *specified* current information requirement also covers the *specified* successor questions. The two requirements may coincide, but that relationship must be established for the task in question.

We use *Coordination Information Theory* (CIT) for a task-relative study of information interfaces: what a receiving task needs, which source distinctions can be removed before optimisation, how the required information is generated, what resolution finite evidence supports, and when composition preserves the resulting contract. *Recursive Coordination Information Theory* (RCIT) addresses those requirements under action-conditioned consequences and repeated interaction. This contract-centred perspective is complementary to coordination-capacity theory, which characterises communication needed to generate prescribed joint distributions (Cuff et al., 2010).

Two companion papers address the static questions. Task-Relative Information Contracts (TRIC) studies criterion-preserving interfaces, their generation, evidential support and conservative hierarchical preservation (Armstrong, 2026a). Task-Sufficient Contraction (TSC) studies source reductions fixed by a declared task before the downstream rate, distortion target, codebook or other operating point is selected (Armstrong, 2026b). Its regret-complete source supplies one possible current-decision contract. The present paper asks whether such a contract also determines action-conditioned successor answers, and what supplementation is required when it does not.

Here “static” means that the contract and its validity conditions are fixed; a system serving it may still learn or receive feedback. A current interface can remain correct for its original role while omitting information required for successor evaluation. A fixed loss and transition mechanism already suffice for this separation. RCIT therefore concerns the additional obligations placed on an interface, rather than an instruction to retain the full state whenever a system acts.

Contribution and approach. The paper develops a contract-to-contract analysis. A family of current queries defines the decision interface. Pulling successor-evaluation queries back through the action law defines a second interface on the same rich state. Factorisation determines when the first interface supplies the second, and conditional entropy quantifies the missing finite deterministic label. Iterating controlled prediction gives an exact recursive completion. Applying the same construction after joining local contracts identifies additional information about their joint evolution.

The analysis combines statistical sufficiency and quotient factorisation with the classical coarsest controlled-Markov refinement (Blackwell, 1953; Bahadur, 1954; Dean and Givan, 1997; Givan et al., 2003). Operator-relative completion also has an established formulation in abstract interpretation (Ranzato and Tapparo, 2007). The contribution is the explicit relationship among current, successor and joint requirements, the information accounting relative to the contract already supplied, and the exact consequences of those distinctions.

The finite constructions include a same-task separation, an eight-state example with two current-regret classes and four closed classes, a multi-stage refinement example, potential-shaping invariance, and a joint-contract witness with positive supplemental information despite zero individual closure costs. Finite rational-inattention geometry supplies a further connection to the Nash-welfare representation of a current information-regret problem (Armenter et al., 2024). Separate recursive-learning investigations examine training and evaluation routes through a factorial comparison, a post-hoc persistence audit and a prospectively frozen target ablation.

2 Static contracts and the recursive question

2.1 A criterion-preserving interface

For a raw state X , a named criterion Y and a deterministic interface $T = t(X)$, TRIC defines

$$D_{\text{TRI}}(T; Y) = I(Y; X \mid T). \quad (1)$$

Zero deficit preserves the full posterior of the criterion. A canonical representative is

$$S_Y(X) = P(Y \in \cdot \mid X). \quad (2)$$

Under the stated regularity conditions, every deterministic zero-deficit interface determines S_Y . For finite variables it consequently has minimum entropy among those interfaces. It preserves attainable Bayes decision value for every downstream loss on the same criterion under the same joint law, using classical sufficiency and decision-equivalence results (Blackwell, 1953; Bahadur, 1954).

TRIC additionally separates loss during reporting from loss during aggregation, studies what resolution a finite evidence budget can support, and gives conditions for conservative internal refinement to preserve a parent-facing contract (Armstrong, 2026a). These questions concern both the information required and the conditions under which a protocol can supply it. The present paper uses Equation (1) to assess a supplied contract against an additional, action-conditioned target.

2.2 A task-certified source for current decisions

TSC asks when a declared task fixes a source reduction that preserves the complete downstream optimisation family, before an operating point is selected (Armstrong, 2026b). Its certification framework distinguishes exact task sufficiency, exactness under architectural restriction, approximation and corrected reductions. The consumer-specific regret source below is one construction within that framework.

For finite source and action sets and finite conditional risks, a consumer $\mathcal{C} = (\mathcal{A}, \ell)$ defines

$$r_x(a) = \mathbb{E}[\ell(a, Y) \mid X = x], \quad b(x) = \min_{a \in \mathcal{A}} r_x(a), \quad g_x(a) = r_x(a) - b(x). \quad (3)$$

The regret-complete source $G_{\mathcal{C}}$ merges x, x' exactly when $g_x(a) = g_{x'}(a)$ for every available action. It retains the cost of every possible mistake while discarding action-independent loss baselines. With a unique optimal action, the matched posterior, regret and optimal-action objects satisfy

$$S_Y \longrightarrow G_{\mathcal{C}} \longrightarrow B_{\mathcal{C}}. \quad (4)$$

The arrows are deterministic coarsenings: S_Y retains the criterion posterior, $G_{\mathcal{C}}$ the complete relative action costs, and $B_{\mathcal{C}}$ the optimum. This is a matched-consumer hierarchy, not a mandatory pipeline for constructing a TSC source.

The finite TSC result states

$$R_X(\delta) = R_{G_{\mathcal{C}}}(\delta), \quad R_X(\delta) = \inf_{p(a|x): \mathbb{E}g_X(A) \leq \delta} I(X; A). \quad (5)$$

Thus the same source preserves the complete one-step rate-regret curve. The minimality of $G_{\mathcal{C}}$ is pointwise preservation of every action regret; it need not be the smallest statistic that happens to preserve the optimised curve. For example, rows $(0, 1)$ and $(0, 2)$ differ pointwise but share a constant zero-regret action. We use the complete current-regret requirement to ask whether successor insufficiency can persist even after all such current comparisons have been retained.

2.3 Scope: characterising and supplying a contract

The formal analysis characterises information requirements for declared queries and a specified model. For a known finite transition kernel it also constructs the coarsest controlled-Markov completion. Learning that kernel, identifying the relevant queries, acquiring observations and implementing the interface in a distributed system are distinct problems. The entropy quantities describe labels under a declared distribution, conditional on information already available; storage, sensing, communication and computation require their own resource models.

A reactive component may receive a newly computed current contract at every step without maintaining a predictive state of its own. The additional requirements studied here arise when the contract must also answer declared successor queries or support recursive prediction. This leaves the original static guarantee intact: the interface is being assessed for a further role. Section 10 develops the resulting questions about learning, evidence and distributed access.

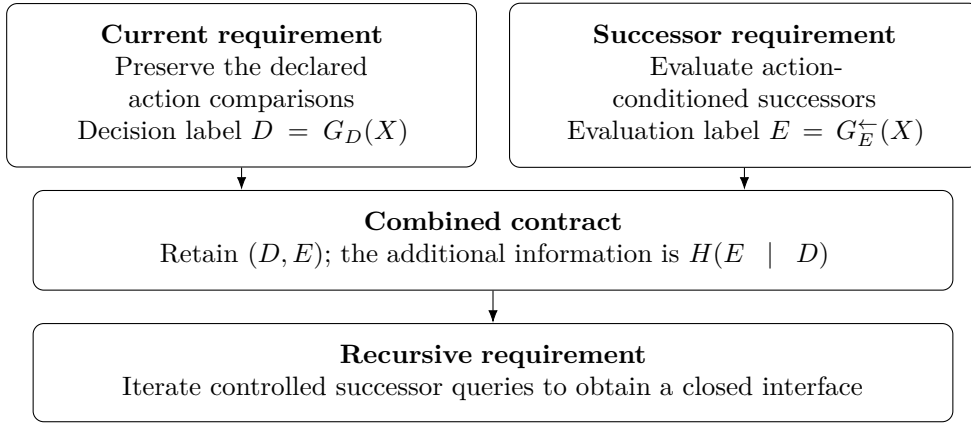


Figure 1: The questions compared by RCIT. Both D and E are functions of the current rich state; the latter is defined by pulling successor queries back through the action law. The arrows combine requirements and then extend them to recursive prediction. Sections 3 and 4 give the exact constructions.

3 Decision and action-conditioned consequence contracts

3.1 Finite query interfaces

Except where explicitly stated, $\mathcal{X}$ is a finite rich-state set and μ is a distribution on it. A query is a deterministic function $q : \mathcal{X} \rightarrow \mathcal{Y}_q$. Its answer may be a label, a scalar, a vector or a probability law. A declared family $\mathcal{Q}$ specifies which answers the receiving task must recover. Define

$$x \sim_{\mathcal{Q}} x' \iff q(x) = q(x') \text{ for every } q \in \mathcal{Q}, \quad G_{\mathcal{Q}}(x) = [x]_{\sim_{\mathcal{Q}}}. \quad (6)$$

A deterministic interface $T = t(X)$ is $\mathcal{Q}$ -sufficient when every query factors through T . The finite setting avoids measurable-selection and quotient-space issues. Those issues require additional hypotheses in a continuous-state extension.

Proposition 3.1 (Query quotient and refinement). *$G_{\mathcal{Q}}$ is $\mathcal{Q}$ -sufficient. Every deterministic $\mathcal{Q}$ -sufficient interface T determines $G_{\mathcal{Q}}$. Enlarging the query family refines its quotient, and the quotient of $\mathcal{Q}_1 \cup \mathcal{Q}_2$ is equivalent to the joint label $(G_{\mathcal{Q}_1}, G_{\mathcal{Q}_2})$.*

Proof. Each query is constant on every equivalence class. Conversely, $T(x) = T(x')$ and sufficiency imply equality of every query at x, x' , hence equality of their class labels. Adding queries can only split classes. Agreement on a union is exactly agreement on both constituent families. $\square$

This factorisation construction underlies sufficient reductions and task-relative quotients (Bahadur, 1954; Bruhacs, 2026). It makes the information required by each declared query family explicit.

For interfaces on a common state set, write $D \preceq E$ when E is finer, meaning $D = h(E)$ for some map h . Their join $D \vee E$ is the common refinement represented by (D, E) . All entropy quantities are in *bits* unless the natural-log convention is explicitly introduced in Section 7.

3.2 Current choice and successor evaluation

Let $\mathcal{Q}_D$ be the queries required by the declared current consumer, and set $D = G_D(X) = G_{\mathcal{Q}_D}(X)$. Choosing an optimal action may need only B_C . Preserving the relative cost of every feasible action instead requires G_C . Risk-sensitive or constrained consumers may require other queries. “Decision contract” does not fix one universal strength of preservation.

An action a in a common finite action set $\mathcal{A}$ produces a successor law $K_a(\cdot | x)$ on a successor state space. Let $\mathcal{E}$ be declared evaluation functionals on such laws. Examples include expected successor loss, the probability of a safety label, or the complete law of a successor regret profile. Pull each functional back to the current state:

$$q_{a,e}^E(x) = e(K_a(\cdot | x)), \quad \mathcal{Q}_E^{\leftarrow} = \{q_{a,e}^E : a \in \mathcal{A}, e \in \mathcal{E}\}. \quad (7)$$

The corresponding label is $E = G_E^{\leftarrow}(X)$. The superscript records that the questions concern successors but the interface is a function of the current rich state. A variable action menu must itself be represented or conditioned upon. The present finite formulation uses a common menu.

For a fixed deterministic decision rule $\pi_D : \text{im } G_D \rightarrow \mathcal{A}$, selected-action evaluation uses only

$$q_e^{E, \pi_D}(x) = e(K_{\pi_D(G_D(x))}(\cdot | x)). \quad (8)$$

Retaining the current decision label and every action’s evaluation answers always determines the selected-action answers. Thus

$$(D, E_{\pi_D}) \preceq (D, E_{\text{all}}), \quad H(E_{\pi_D} | D) \leq H(E_{\text{all}} | D). \quad (9)$$

The conditioning on D matters: an all-action evaluation label alone need not specify which action the decision rule selected.

Proposition 3.2 (Transfer from decision to consequence evaluation). *The current decision contract is sufficient for the declared pulled-back evaluation family if and only if*

$$\boxed{G_E^{\leftarrow} = h(G_D)} \quad (10)$$

for some deterministic h . The joint contract is $G_D \vee G_E^{\leftarrow}$.

Proof. If Equation (10) holds, every evaluation query factors through G_D . Conversely, apply proposition 3.1 to the evaluation family with G_D as its sufficient interface. $\square$

Factorisation therefore identifies when the information supplied for current choice also supplies the declared consequence answers.

Prediction is another declared role. A planner may require trajectory distributions, values under a family of future policies, or probabilities of future events. These generate another query family. There is no universal order among a decision interface, an evaluation interface and a prediction interface. An order follows only when one requested family is contained in, or recoverable from, another. Full physical prediction may exceed a declared evaluator’s needs. Conversely, a final-action label may be inadequate for the search procedure that constructs the candidate set (Wei et al., 2026).

Successor laws and realised observations. The interface in Equation (7) concerns a known action-conditioned law. It determines the declared properties of that law, rather than which random successor will occur. Auditing a realised consequence may therefore require observing that consequence. This distinction fixes the target of the evaluation contract.

3.3 A same-task separation

Use a fixed action set $\{a, b\}$ and a single state-dependent loss function on four states. The complete loss and regret rows coincide in this construction because each row has minimum zero:

state	$L(s, a)$	$L(s, b)$	successor under a
x_1	0	1	y_1
x_2	0	1	y_2
y_1	0	1	y_1
y_2	1	0	y_2

(11)

Let action b leave each state unchanged. Those transitions complete the model but are not needed for the separation. In both current states x_1, x_2 , the unique optimal one-step action is a and the complete current regret profile is $(0, 1)$. Following a produces different next regret profiles under the *same* loss: $(0, 1)$ at y_1 and $(1, 0)$ at y_2 .

Proposition 3.3 (Complete current regret need not determine successor regret). *There is a finite, fixed-task decision system in which the complete current feasible-action regret interface is exact, yet it is insufficient to recover the successor regret profile produced by the common unique current optimum. With the uniform law on x_1, x_2 , the additional evaluation label contains one bit conditional on the current contract.*

Proof. The current regret interface merges x_1, x_2 . The selected-action successor query separates them. Hence no map of the current interface recovers that query. Under the specified law, the current label is constant and the evaluation label has two equiprobable values. $\square$

The separation holds with every current action regret retained and with the loss unchanged. Its two roles are complete one-step comparison and evaluation of the next decision geometry. A fully specified long-horizon Bellman contract would impose its own consequence requirements; this example concerns the transfer from the stated one-step contract.

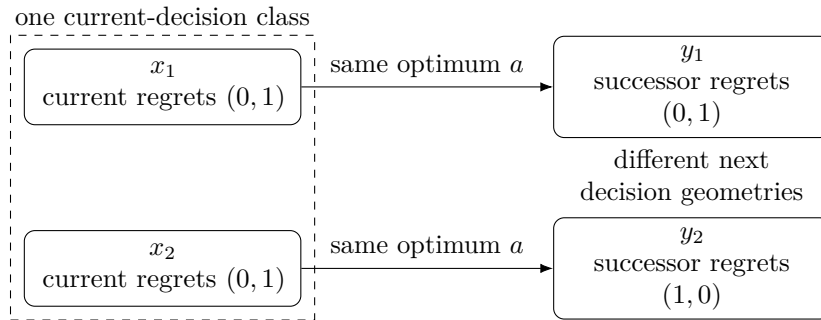


Figure 2: An exact separation under one fixed loss and action set. Uniformly choosing between the two current states gives $\Delta_{E|D} = 1$ bit.

3.4 Residual information and the connection back to TRIC

Definition 3.4 (Residual consequence information). For finite deterministic labels under μ , define

$$\Delta_{E|D} = H_\mu(E | D). \quad (12)$$

Theorem 3.5 (Minimum conditional information in a deterministic supplement). *Among deterministic supplements $Z = f(X)$ such that (D, Z) recovers the declared evaluation answers,*

$$\boxed{\min_Z H_\mu(Z | D) = H_\mu(E | D).} \quad (13)$$

Moreover, $\Delta_{E|D} = 0$ exactly when E is a function of D almost surely. If μ has full support, this is equivalent to pointwise transfer on $\mathcal{X}$.

Proof. An admissible (D, Z) determines the coarsest evaluation label E . Therefore $H(E | D) \leq H(Z | D)$. Taking $Z = E$ attains equality. The zero condition is the standard finite conditional-entropy criterion for deterministic recoverability. $\square$

The contract construction identifies the label being accounted for. The minimum is conditional on the supplied decision information D , rather than a minimum unconditional storage size; its operational scope is stated in Section 2.3.

The connection to TRIC is exact. Because E is a deterministic function of X ,

$$\boxed{D_{\text{TRI}}(D; E) = I(E; X | D) = H(E | D) = \Delta_{E|D}.} \quad (14)$$

TRIC supplies the deficit measure, TSC can supply an exact current-decision contract, and the action-conditioned construction supplies the successor target against which that contract is now assessed. The same deficit thus connects the static contract to its action-conditioned evaluation target.

3.5 Lossy supplementation

If a decoder already knows D and must reconstruct E , an ordinary conditional rate–distortion problem gives

$$R_{E|D}(\delta) = \inf_{p(\hat{e}|e,d): \mathbb{E}d_E(E, \hat{E}) \leq \delta} I(E; \hat{E} | D). \quad (15)$$

For finite reconstruction alphabets and a distortion that is zero exactly on the diagonal,

$$R_{E|D}(0) = H(E | D) = \Delta_{E|D}. \quad (16)$$

An encoder observing the richer X gains nothing over (D, E) for this stated family: averaging its output channel within (D, E) fibres preserves the distortion and cannot increase conditional mutual information. A reduced channel pulls back to X . This is another application of the contraction argument, now to the residual target.

This is the conditional rate–distortion problem with D supplied as side information (Gray, 1972; Cover and Thomas, 2006). A coding interpretation additionally specifies block structure and permissible reconstruction error. Reconstructing an exact closure label approximately and constructing an interface with approximately closed dynamics are different problems; Section 10 addresses the latter.

4 Recursive completion of a task contract

The one-step consequence interface may itself be inadequate for the next step. An exact finite controlled-Markov model makes this requirement explicit. Let $S_t \in \mathcal{S}$, let $\mathcal{A}$ be a common finite action set, and let $P_a(s' | s)$ be a known transition kernel. An MDP additionally specifies rewards or costs. The initial contract $D_0 : \mathcal{S} \rightarrow \mathcal{Z}_0$ declares which state distinctions must be retained. It may be a current-regret profile, a safety label, a posterior contract or a joint requirement.

For an interface D , define its successor signature

$$\mathcal{R}_P(D)(s) = \left(D(s), \left\{ \sum_{s': D(s')=z} P_a(s' | s) \right\}_{a \in \mathcal{A}, z \in \text{im } D} \right), \quad (17)$$

with identical signatures assigned the same label. The interface is *controlled-Markov* when the probabilities of its next labels depend only on its current label and the action. Equivalently, $\mathcal{R}_P(D)$ is the same partition as D .

Starting with D_0 , iterate $D_{k+1} = \mathcal{R}_P(D_k)$. Every strict step increases the number of classes, so the sequence stabilises after at most $|\mathcal{S}| - |D_0|$ strict steps. Denote the fixed point by

$$D_\infty = \text{Cl}_P(D_0). \quad (18)$$

This homogeneous-partition refinement applies to any starting partition (Dean and Givan, 1997; Givan et al., 2003). We use it to complete the declared task contract and account for the information added.

Theorem 4.1 (Coarsest controlled-Markov refinement (Dean and Givan, 1997; Givan et al., 2003)). *D_∞ is the unique coarsest controlled-Markov refinement of D_0 , up to relabelling. In particular, every controlled-Markov interface U with $D_0 \preceq U$ satisfies $D_\infty \preceq U$.*

Proof. Refinement and stability follow from construction. For minimality, suppose U is stable and $D_k \preceq U$. Each D_k -class is a union of U -classes. Two states with the same U label have identical transition probabilities into every U -class. Summing over those contained in each D_k -class gives identical successor signatures. Hence $D_{k+1} \preceq U$. Starting from $D_0 \preceq U$, induction gives $D_\infty \preceq U$. Any two coarsest stable refinements therefore determine each other. $\square$

4.1 What the completed interface preserves

The stable partition defines an exact abstract kernel

$$\bar{P}_a(z' | z) = \sum_{s': D_\infty(s')=z'} P_a(s' | s), \quad D_\infty(s) = z. \quad (19)$$

The value is independent of the representative s . For a policy $\pi_t(a | s) = \bar{\pi}_t(a | D_\infty(s))$, the projected process has exactly this kernel and policy. Induction on a path's length gives equality of its finite-dimensional distributions in the raw and abstract models. Since D_0 is a function of D_∞ , the quotient reproduces every bounded finite-horizon query of

$$(D_0(S_0), A_0, D_0(S_1), A_1, \dots, D_0(S_H)) \quad (20)$$

under these policies and the corresponding projected initial law.

The task-process guarantee covers policies implemented through D_∞ , rather than policies that select different actions inside an abstract class. Preserving absolute or cumulative reward additionally requires the relevant reward quantities to factor through the retained state, or a separate preservation argument. A current-regret seed alone therefore does not guarantee the original absolute-return problem.

The construction also differs from minimising an arbitrary history-based predictive representation. It requires a deterministic function of the current rich state with exact one-step controlled-Markov dynamics. Information states, predictive-state representations and trace equivalences have related but not interchangeable preservation requirements (Subramanian et al., 2022).

4.2 Information required beyond the starting contract

Definition 4.2 (Recursive closure information). For a fixed state law μ , define

$$\boxed{C_\mu(D_0) = H_\mu(D_\infty \mid D_0)}. \quad (21)$$

Corollary 4.3 (Minimum deterministic supplement for closure). *Among deterministic $Z = f(S)$ such that (D_0, Z) is controlled-Markov,*

$$C_\mu(D_0) = \min_Z H_\mu(Z \mid D_0) = \sum_{k=0}^{m-1} H_\mu(D_{k+1} \mid D_k), \quad (22)$$

where $D_m = D_\infty$. Also

$$0 \leq C_\mu(D_0) \leq H_\mu(S \mid D_0). \quad (23)$$

Proof. Every admissible (D_0, Z) determines D_∞ by Theorem 4.1. Conditional entropy gives the lower bound on the supplement, attained by $Z = D_\infty$. For nested partitions, $H(D_{k+1} \mid D_k) = H(D_{k+1}) - H(D_k)$, so the sum telescopes. Finally, D_∞ is a function of S . $\square$

The first layer $H(D_1 \mid D_0)$ equals the one-step residual for predicting the next D_0 label under every action. Later layers arise because the distinctions restored at one step may themselves require further distinctions for prediction. These are rounds of a refinement procedure. They are not automatically a universal measure of planning horizon.

The kernel and base contract determine the closure partition. The distribution μ determines its information content. It may be a specified initial-state distribution or a deployment distribution, but must be declared. If μ has full support, zero closure information is equivalent to D_0 already being stable. Without full support, a zero value means agreement only almost surely under the accounting law. It does not certify states excluded from that law, including states that later actions may reach.

Equation (14) applies again:

$$C_\mu(D_0) = D_{\text{TRI}}(D_0; D_\infty). \quad (24)$$

The completion is a new evaluation target for the starting contract.

4.3 Structure of the completion

The map Cl_P is extensive, monotone and idempotent. If $D \preceq E$, any stable refinement of E is a stable refinement of D , so

$$\text{Cl}_P(D) \preceq \text{Cl}_P(E), \quad \text{Cl}_P(\text{Cl}_P(D)) = \text{Cl}_P(D). \quad (25)$$

Thus a matched static hierarchy lifts to a hierarchy of dynamic contracts. A strict static difference need not remain strict after completion.

There is an equivalent operator statement. Let $\mathcal{A}(D)$ be the algebra of real functions constant on D -classes, and let $(P_a f)(s) = \sum_{s'} P_a(s' \mid s) f(s')$. Then

$$D \text{ is controlled-Markov} \iff P_a \mathcal{A}(D) \subseteq \mathcal{A}(D) \quad \forall a. \quad (26)$$

The forward direction follows by summing over blocks. The reverse follows by choosing block indicators. Consequently, $\mathcal{A}(D_\infty)$ is the smallest invariant finite partition algebra containing $\mathcal{A}(D_0)$. Closure of the algebra, not only a list of scalar summaries, matters. This is the finite partition-algebra formulation of operator-relative completion, whose general abstraction framework is developed by Ranzato and Tapparo (2007).

Restricting the supported action family can reduce the completion. If $U_1 \subseteq U_2 \subseteq \mathcal{A}$, then $\text{Cl}_{P,U_1}(D_0) \preceq \text{Cl}_{P,U_2}(D_0)$. A fixed D_0 -measurable policy similarly defines a single supported kernel. The retained state therefore depends on whether the contract must support that policy, a subset of interventions or all counterfactual actions.

Although the total closed representation is monotone in the starting task, its *additional* information cost need not be. A constant contract is already stable. Refining it to a non-stable label can make C_μ positive. Further refining to its exact closure makes the remaining cost zero. Reporting only a scalar “amount of information required” can obscure which initial contract has already been supplied.

5 Exact task-state reductions and shaping invariance

5.1 An intermediate eight-state quotient

A finite MDP consists of states, actions, a controlled transition law and a reward or cost. The raw state is adequate for the model by construction, but an interface need not retain every raw distinction for every task. This example asks only for exact prediction of the current-regret process under quotient-implementable control.

There are eight states $p_1, \dots, p_4, q_1, \dots, q_4$ and actions a, b . The immediate rewards and deterministic transitions are completely specified in Table 1. Define current regret as $g_s(c) = \max_{u \in \{a,b\}} r(s, u) - r(s, c)$. All p states have row $(0, 1)$, all q states row $(1, 0)$, so the initial contract has two classes.

State	$r(s, a)$	$r(s, b)$	next under a	next under b
p_1	10	9	p_1	q_1
p_2	4	3	p_1	q_1
p_3	6	5	q_3	p_3
p_4	6	5	q_3	p_3
q_1	9	10	p_3	q_1
q_2	3	4	p_3	q_1
q_3	5	6	q_3	p_1
q_4	5	6	q_3	p_1

Table 1: Complete eight-state construction. The task seed retains immediate relative action rewards, not their absolute levels.

The first successor refinement separates the two transition types inside each initial class:

$$D_1 = \{p_1, p_2\} \mid \{p_3, p_4\} \mid \{q_1, q_2\} \mid \{q_3, q_4\}. \quad (27)$$

Each block now has a well-defined successor block under each action, so $D_\infty = D_1$. By contrast, reward-respecting bisimulation must distinguish p_1 from p_2 and q_1 from q_2 , while retaining the identical pairs p_3, p_4 and q_3, q_4 . Write that partition as B_R . The exact cardinalities are

$$|D_0| = 2 < |D_\infty| = 4 < |B_R| = 6 < |\mathcal{S}| = 8. \quad (28)$$

Under the uniform accounting law, their entropies are 1, 2, 2.5, 3 bits. Hence completing the current task contract requires one additional bit, compared with two bits to restore the raw state. These quantities are computed exactly from the displayed model.

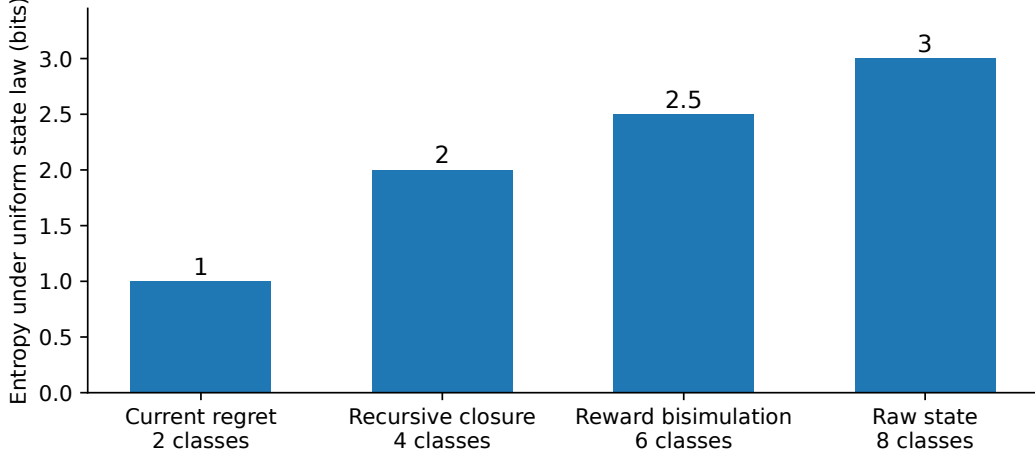


Figure 3: Information retained by four distinct contracts in the eight-state construction under a uniform state law. The recursive completion adds one bit to the current-regret contract. Reward-preserving abstraction and raw state retain additional information because they answer richer questions.

States p_1 and p_2 merge in D_∞ despite different absolute rewards. The quotient preserves the declared regret-label process, whereas reward-respecting bisimulation also retains absolute reward levels. The class counts therefore compare these different preservation requirements; they do not establish improved control of the original absolute-return problem.

5.2 One successor check can be insufficient

A separate one-action chain has states $s_0, \dots, s_4$, with $s_0 \rightarrow s_0$ and $s_i \rightarrow s_{i-1}$ for $i > 0$. Start with the label distinguishing s_0 from every other state. Refinement successively separates s_1 , then s_2 , then s_3 from s_4 :

$$|D_0| = 2 \longrightarrow |D_1| = 3 \longrightarrow |D_2| = 4 \longrightarrow |D_3| = 5. \quad (29)$$

The first-step answers do not determine the second-step distinction. Under the uniform law, the increments are approximately 0.6490225, 0.5509775 and 0.4 bits, adding to 1.6 bits. Appendix A gives the full partitions and checks. The example illustrates the difference between evaluating the next contract and carrying a representation closed under repeated prediction.

5.3 Potential-based reward shaping

A different decision contract is appropriate when the current comparison uses optimal discounted values. In a finite discounted MDP with $0 < \gamma < 1$ and bounded rewards, define

$$g^*(s, a) = V^*(s) - Q^*(s, a), \quad G^*(s) = (g^*(s, a))_{a \in \mathcal{A}}. \quad (30)$$

This is optimal deviation regret, the negative of optimal advantage. It is not the immediate-regret seed of the preceding example.

Potential-based reward shaping replaces R by

$$R^\phi(s, a, s') = R(s, a, s') + \gamma\phi(s') - \phi(s) \quad (31)$$

for a bounded potential ϕ . Classical shaping invariance gives $Q^{\pi, \phi}(s, a) = Q^\pi(s, a) - \phi(s)$ and $V^{\pi, \phi}(s) = V^\pi(s) - \phi(s)$, so $G^{*, \phi} = G^*$ (Ng et al., 1999). The discounted telescoping proof and terminal qualification are included in Appendix A.4.

Corollary 5.1 (Closure of an invariant seed). *With the kernel and action set unchanged by Equation (31), every refinement level seeded by G^* is unchanged up to relabelling. In particular,*

$$\text{Cl}_P(G^{*,\phi}) = \text{Cl}_P(G^*). \quad (32)$$

The closure information is also unchanged when evaluated under the same state law.

Proof. The starting partitions and kernels coincide. Formula (17) therefore yields the same partition at every subsequent step. $\square$

Shaping invariance and refinement together preserve the complete task interface: a transformation irrelevant to the decision geometry leaves its recursive completion unchanged.

An exact family separates that contract from absolute-reward preservation without a uniform finite bound. Take states p_i, q_i , $1 \leq i \leq m$, with both actions self-loops, rewards $(1, 0)$ at each p_i and $(0, 1)$ at each q_i . Let $\phi(p_i) = i$ and $\phi(q_i) = m + i$. The two optimal-regret rows remain $(0, 1)$ and $(1, 0)$, and their two-class partition is already closed. The shaped immediate reward vectors are all distinct, so shaped reward-respecting bisimulation has $2m$ classes. Under the uniform law,

$$H(B_{R^\phi} \mid \text{Cl}_P(G^*)) = \log_2 m. \quad (33)$$

Every individual finite model has bounded potential; the family varies with m . The unbounded gap concerns exact preservation and does not imply the same comparison for approximate reward models. Computing G^* may itself require solving the original decision problem, so the quotient size and the cost of constructing its seed remain separate.

6 Joint contracts and coordination-induced information

Suppose several agents or components require different projections of one common controlled state. All interfaces below refer to the *same* global kernel and supported action family. For each initial task D_i , let $C_i = \text{Cl}_P(D_i)$. Each local contract is already exact for its own next-label distribution. Their joint current interface is

$$J = \bigvee_{i=1}^n C_i. \quad (34)$$

Knowing every correct next marginal does not determine the joint next distribution. As a result, J need not be controlled-Markov even when every C_i is.

Definition 6.1 (Coordination-induced closure information).

$$\boxed{\Gamma_\mu(D_1, \dots, D_n) = H_\mu(\text{Cl}_P(J) \mid J)}. \quad (35)$$

This is the supplemental deterministic state information needed to close the joined contract after the local completions have already been supplied. Its minimum-supplement interpretation follows directly from corollary 4.3. Equivalently, $\Gamma_\mu = D_{\text{TRI}}(J; \text{Cl}_P(J))$. The quantity measures a global state-information requirement. An inter-agent communication interpretation requires the observation and message model discussed below.

6.1 An exact five-state witness

Consider the one-action Markov chain

$$P = \begin{pmatrix} 0 & 1/3 & 1/3 & 1/6 & 1/6 \\ 1/4 & 1/4 & 1/2 & 0 & 0 \\ 0 & 1/2 & 1/4 & 1/4 & 0 \\ 1/3 & 1/6 & 1/6 & 0 & 1/3 \\ 1/2 & 0 & 0 & 1/4 & 1/4 \end{pmatrix} \quad (36)$$

on states 0, 1, 2, 3, 4. Define

$$C_1 = \{0, 1, 3\} \mid \{2, 4\}, \quad C_2 = \{0, 2, 3\} \mid \{1, 4\}. \quad (37)$$

For C_1 , states in its first block have probability 1/2 of returning to that block, and states in the second have probability 3/4 of entering the first. The same sums hold for C_2 with its corresponding blocks. Thus both are lumpable.

Their join is

$$J = \{0, 3\} \mid \{1\} \mid \{2\} \mid \{4\}. \quad (38)$$

From state 0, the probability of entering the joint block $\{0, 3\}$ is 1/6. From state 3, it is 1/3. The join therefore fails stability, and one refinement makes it the raw-state partition. Under the uniform law, its only unresolved block has probability 2/5 and contains one fair bit:

$$C_\mu(C_1) = C_\mu(C_2) = 0, \quad \Gamma_\mu(C_1, C_2) = \frac{2}{5} \text{ bit}. \quad (39)$$

The local marginals are exact. The omitted distinction concerns their joint successor behaviour. In the language of Markov aggregation, this is a failure of joint lumpability; Equation (39) quantifies its repair relative to the supplied joint label.

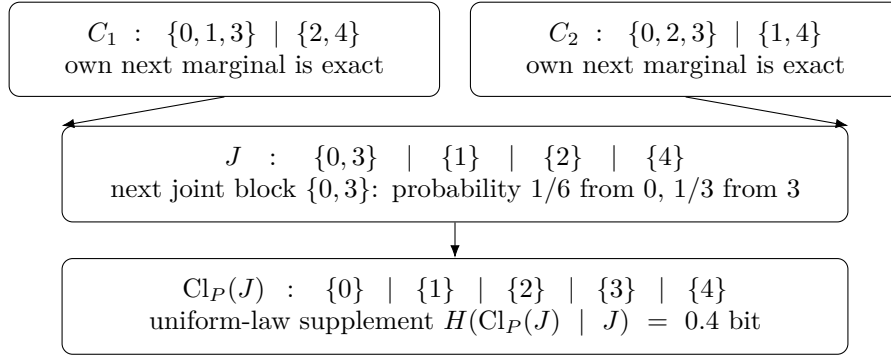


Figure 4: Joining two locally closed interfaces exposes a distinction absent from their joint current label. The deficit concerns the joint next law, not either local marginal.

6.2 Products and the size of the joint requirement

Proposition 6.2 (Product amplification). *Take k independent copies of Equation (36), with the product-uniform state law. The product interfaces $C_1^{\otimes k}$ and $C_2^{\otimes k}$ remain individually closed, whereas*

$$\Gamma_k = k\Gamma_1 = \frac{2k}{5} \text{ bits}. \quad (40)$$

Proof. For a product partition $D^{\otimes k}$ and product kernel, the next product-block probabilities are products of the coordinate next-block probabilities. Equality of all product probabilities implies equality of their coordinate marginals, and equality of the marginals implies equality of the product. Therefore one refinement of the product partition is the product of the coordinate refinements. Applying this observation repeatedly proves that product completion equals the product of the individual completions. In the present witness, each C_i is already stable, while the completion of J is the raw five-state partition. Conditional entropy is additive across independent coordinates under the product state law, giving Equation (40). $\square$

The amplification is a property of independent replicas with the specified product interfaces. It is not a claim of arbitrary additivity under coupled dynamics or correlated source laws.

There is also a useful general identity, inherited from closure-operator properties:

$$\text{Cl}_P \left(\bigvee_i D_i \right) = \text{Cl}_P \left(\bigvee_i \text{Cl}_P(D_i) \right). \quad (41)$$

The left side is a stable refinement of every D_i , so it refines each local closure and their join. Conversely, the join of local closures refines the join of original tasks. Monotonicity and idempotence give equality. Closing the constituents first does not lose the ability to reach the same joint completion, but it does not eliminate the possible final joint deficit.

6.3 Relation to conservative hierarchy and distributed control

TRIC’s conservative-substitution result and the present non-composition example have different premises. TRIC preserves an already sufficient parent interface under conditions shielding the parent and preserving its external dependence structure. Here the kernel is fixed, and the requested contract changes from separate next marginals to their joint next law. The two guarantees therefore concern different requirements at the receiving boundary.

A condition such as conditional independence of the next local labels given the current rich state and action is sufficient for the local next marginals to determine the joint next law, provided those marginals already factor through the local contracts. It is not necessary: other known dependence structures can also permit composition. The harder operational question is how distributed components acquire, represent or communicate the missing dependence when no such condition is available. Existing decentralized information-state theory already addresses compressed shared and local information for control (Subramanian et al., 2022). The proposed Γ problem must be specialised to a particular observation and communication model before it can give an operational rate or a protocol.

7 Current decision geometry as an information–welfare problem

The preceding sections concern which questions an interface can answer. A fixed current regret table also induces an information–performance tradeoff. That tradeoff has a useful welfare representation, but it is not the whole recursive contract.

Units and domain. In this section only, information and entropy use *natural logarithms*. Let $p(s) > 0$ on a finite state set, $\sum_s p(s) = 1$, let $\mathcal{A}$ be finite, and let $d(s, a) \geq 0$ be finite regrets with at least one zero in every row. Define

$$F_\beta = \min_{Q(a|s)} \{I_{\text{nat}, Q}(S; A) + \beta \mathbb{E}_Q d(S, A)\}, \quad \beta \geq 0. \quad (42)$$

States of zero probability can be removed before this optimisation. This operation is separate from full-domain structural closure.

For a common action mixture $\pi \in \Delta(\mathcal{A})$, let

$$u_s(\pi; \beta) = \sum_a \pi(a) e^{-\beta d(s, a)}, \quad \text{NSW}_\beta^* = \max_\pi \prod_s u_s(\pi; \beta)^{p(s)}. \quad (43)$$

These are transformed regret utilities. They are not the original economic, network or task utilities.

Proposition 7.1 (Nash-product form of the fixed-slope problem). *Under the finite assumptions above,*

$$\boxed{F_\beta = -\log \text{NSW}_\beta^*}. \quad (44)$$

Proof. For a channel Q with action marginal Q_A ,

$$\sum_s p(s) D_{\text{KL}}(Q(\cdot | s) \| \pi) = I_{\text{nat}, Q}(S; A) + D_{\text{KL}}(Q_A \| \pi).$$

Minimising over π recovers the mutual information. For fixed π , minimising over each row $Q(\cdot | s)$ gives the Gibbs row proportional to $\pi(a)e^{-\beta d(s,a)}$, with minimum $-\log u_s(\pi; \beta)$. Minimising the weighted sum of these values and exponentiating gives Equation (44). Zero mixture entries are handled by restricting the row to the mixture support. $\square$

The variational argument is classical rate–distortion machinery (Blahut, 1972; Cover and Thomas, 2006). More directly, Armenter et al. (2024), in a construction available as a 2021 working paper, study the same finite optimisation through exponentiated payoff vectors and a prior-weighted logarithmic objective. To see the exact mapping, set the rational-inattention payoff to $v(s, a) = -d(s, a)$ and the information price to $\lambda = 1/\beta$ for $\beta > 0$. Then

$$\max_Q \{ \mathbb{E}_Q v - \lambda I_{\text{nat}, Q} \} = -F_\beta / \beta, \quad e^{v(s,a)/\lambda} = e^{-\beta d(s,a)}. \quad (45)$$

The convex-hull geometry, unique utility optimum and support bound are the finite-rational-inattention results of Armenter et al. (2024). Their Nash-product form expresses the current information–regret optimisation as a coordination objective.

For completeness, define

$$c_a^\beta = (e^{-\beta d(s,a)})_s, \quad \mathcal{U}_\beta = \text{conv}\{c_a^\beta : a \in \mathcal{A}\}. \quad (46)$$

The optimum of $\sum_s p(s) \log u_s$ over $\mathcal{U}_\beta$ is unique. Its proportional-fair condition is

$$\sum_s p(s) \frac{u_s - u_s^*}{u_s^*} \leq 0 \quad (u \in \mathcal{U}_\beta). \quad (47)$$

An optimal action mixture need not be unique, but one exists using at most $|\mathcal{S}|$ actions. Appendix B states the supporting conditions and short proofs, attributing their geometry to Armenter et al. (2024) and their proportional-fair interpretation to the weighted-log tradition (Kelly et al., 1998).

Writing the one-step rate–regret function in nats, ordinary convex duality yields

$$R_{\text{nat}}(\delta) = \sup_{\beta \geq 0} \{ -\log \text{NSW}_\beta^* - \beta \delta \}. \quad (48)$$

Thus the family of welfare optima contains the full one-step rate–regret curve. At differentiable slopes, F'_β is the optimal expected regret. With feasible zero regret,

$$\lim_{\beta \rightarrow \infty} \text{NSW}_\beta^* = e^{-R_{\text{nat}}(0)}. \quad (49)$$

Under unique zero-regret actions, $R_{\text{nat}}(0) = H_{\text{nat}}(B_C)$, the familiar optimal-action endpoint also explicit in Walsh (2026b). The curve and endpoint relations follow from duality, envelope and finite-alphabet limiting arguments. If entropy is expressed in bits, the same endpoint is $2^{-H(B_C)}$.

The scalar and the recursive contract. F_β describes the current declared matrix, source law and operating slope. It need not determine the successor evaluation contract, a learning trajectory, or the effect of changing the state-dependent data generator. The recursive experiments below tested and rejected stronger predictive interpretations for several particular scalar and structural summaries. Their failure does not negate the fixed-slope identity. It identifies a limit to using that identity as a general recursive state description.

TACIT uses reputation-weighted Nash welfare to arbitrate among O-RAN rApp proposals with positive utilities (Armstrong et al., 2026). There the weights represent reliability and the utilities represent reported network consequences. In Equation (43), weights are source probabilities and utilities are Gibbs transforms of regret. The product objectives thus aggregate different quantities. Relating their dynamics requires an explicit mapping of agents, utilities, probabilities and updates.

8 Recursive-learning investigations

The experiments investigate implemented information routes in recursive learning, separately from the exact contract constructions. They examine a criterion-coupling factorial comparison, a post-hoc persistence audit and a prospectively frozen target ablation. These comparisons test the effects of particular training and evaluation routes rather than identifying canonical sufficient interfaces. Appendix C gives the protocols, inferential conventions and complete summary tables.

8.1 From scalar prediction to separate interfaces

The initial classification analyses used 160 saved parents, 480 synchronized three-action forks and 20 original units across five fixed lineages. Actions produced different estimated successor geometries in every lineage, and predeclared approximate-current-value witnesses occurred in all five. One count-derived scalar equality, $J = 133/160$ for two actions at parent `13_k00__t16`, coexisted with different estimated child geometry. Importantly, both corresponding exact point partitions had six singleton classes. The observation therefore does not demonstrate that two equal complete decision quotients led to different successors. It is an insufficiency witness for the reported scalar description of the estimated geometry, not for an unknown population quotient (see Appendix C.7).

Induced F , relevant-but-unresolved distinction burden, and higher temporal derivatives failed their declared continuation-prediction gates. These findings motivate the separate assessment of current selection, successor training and external evaluation. They concern the tested summaries, whereas proposition 3.3 concerns exact contracts in a specified model.

8.2 Training and evaluation provenance

The completed factorial investigation has five fixed lineages, four paired original units per lineage and 32 recursive cycles. Six receiver roles evolve within each unit. The primary hidden capability averages the ten fixed class scores and receiver roles A0–A2 equally. A separate external steering panel can score interventions, while a disjoint hidden audit provides the authoritative performance measure. The audit is not used for intervention selection.

The core factorial crosses synthetic-only or mixed gradient-update training with endogenous or external- Y intervention evaluation. Two controls add external evaluation of a misaligned criterion and persistence with the initially selected intervention. The misaligned criterion is the equal-weight capability of classes 3 and 9, rather than the ten-class primary mean. It tests this particular criterion mismatch, not every possible unrelated target. The six conditions are fully listed in Table 2.

The literal three-intervention menu, selection rule, optimiser, correction budget and cadence remain fixed. Both synthetic and mixed conditions retain the 1,024-example synthetic correction. Inherited checkpoints and generators have original-data ancestry. External steering changes intervention selection and therefore influences learning, even when steering examples never enter gradient updates. Its route into learning is intervention selection rather than gradients.

The two primary outcomes are terminal preservation and normalized trajectory area, computed by trapezoidal integration over generations 0–32 divided by 32. The bootstrap resamples

Condition	New base-transition training	Intervention evaluation
SYN-ENDO	6,000 synthetic examples	Endogenous panel
SYN-EXTY	6,000 synthetic examples	External primary-Y panel
MIX-ENDO	5,000 synthetic + 1,000 genuine examples	Endogenous panel
MIX-EXTY	5,000 synthetic + 1,000 genuine examples	External primary-Y panel
SYN-EXTWRONG	6,000 synthetic examples	External classes-3/9 criterion
SYN-NOREVAL	6,000 synthetic examples	Fixed initial intervention

Table 2: Completed criterion-coupling design. Every condition also retains the same 1,024-example synthetic correction step. “Synthetic-only” refers to gradients during these new trajectories, not to the ancestry of inherited checkpoints or generators.

whole original units within each fixed lineage, gives the five lineages equal weight and preserves pairing across conditions and generations. Each primary endpoint uses a 97.5% interval under the frozen two-endpoint convention. Those intervals are not simultaneous guarantees over every contrast or diagnostic.

SYN-ENDO deteriorated by 3.837 pp at the terminal generation, with a descriptive 95% interval of $[-4.368, -3.353]$ pp and deterioration in all five lineage means. External-Y evaluation gave a small trajectory-area benefit over endogenous evaluation, +0.268 pp with primary interval $[0.103, 0.440]$ pp, but its terminal difference was uncertain. Crucially, EXTY and EXTWRONG met the declared equivalence criterion: their terminal and area contrast intervals both lay inside ± 1 pp. Intervention disagreements were substantial, 61.9% for EXTY versus ENDO and 43.4% for EXTY versus EXTWRONG. Thus lack of treatment exposure does not explain the failed specificity result (see Appendix C.4).

The completed classification is CC-B/CC-D, not CC-A. The experiment operationally separates training and evaluation routes, but does not establish criterion-specific external evaluation as the protective mechanism. The much larger mixed-training benefit remains. Full condition summaries and primary contrasts are retained in Appendix C; no unsuccessful arm is omitted.

8.3 Persistence and locally harmful corrections: exploratory analysis

A post-hoc audit compared adaptive interventions with SYN-NOREVAL. Persistence had a +0.306 pp normalized-area advantage over ENDO, with descriptive 95% interval $[0.155, 0.458]$ pp and four of five positive lineage means. The terminal contrast, +0.212 pp with interval $[-0.265, 0.714]$ pp, remained unresolved.

All 20 ENDO trajectories eventually departed from the persistent intervention. At their first divergences, 13 selected applied children had lower hidden-Y capability than the persistent children and seven had higher capability. The mean selected-minus-persistent effect was -0.356 pp, with descriptive interval $[-0.597, -0.112]$ pp. The first-switch audit checked common preceding actions, parent checkpoint and parameter identities, candidate menus and application seeds. Its integrity chain includes direct initial-checkpoint rehashes and the original sealed audit/pruning records for later parents (see Appendix C.5).

The selector’s forecast children used different probe randomness and were not hidden-audited or retained. These results therefore establish locally harmful *applied* corrections in the tested forks, not a reconstructed inversion of the exact forecast children’s hidden scores. Later differences are whole-policy consequences, not isolated mediators. The seven beneficial ENDO switches remain part of the result. This is exploratory evidence consistent with overcorrection in the

fixed cohort, not a theorem that persistence is generally superior or an empirical identification of G_D and $G_E^\leftarrow$.

8.4 What the genuine training examples supply: frozen ablation

Phase B compares TRUE-ENDO (the genuine-label MIX condition), REALX-PSEUDOY-ENDO and SYN-ENDO. PSEUDO uses the identical intended 1,000 genuine input IDs at each unit, role and generation, but replaces their targets with hard argmax predictions from that condition’s current *raw pre-decision parent*. The pseudo-targets are frozen before candidate forecast training. Genuine inputs do not enter the synthetic correction forecasts. They enter the selected branch’s subsequent base transition, and the generator is refitted on the same combined transition data (see Appendix C.2).

The genuine targets are excluded from PSEUDO training and selection. Offline pseudo-label accuracy is assessed only after all treatment trajectories complete and are sealed. Historical TRUE and SYN controls were reused under a protocol requiring exact full-horizon replay before reuse. The completed experiment used historical anchors; Appendix C.2 states the replay requirements and Appendix D states what is released with this version.

Contrast	Terminal preservation (pp)	Normalized area (pp)
TRUE – PSEUDO	8.226 [7.749, 8.668]	7.347 [7.184, 7.516]
PSEUDO – SYN	0.856 [0.337, 1.440]	1.039 [0.857, 1.217]
TRUE – SYN	9.082 [8.502, 9.687]	8.387 [8.244, 8.526]

Table 3: Completed target ablation. Brackets are the reported primary 97.5% intervals. All three contrasts are positive in all five lineage means on both endpoints. These are paired differences, not differences inferred by subtracting rounded condition means.

The frozen classification is B-C: both directed contrasts are beneficial on a common primary endpoint. Their magnitudes are not comparable in size. Retaining genuine targets supplies a substantially larger increment than the positive benefit of genuine inputs carrying self-generated targets. PSEUDO still ends below its initial hidden capability (-2.981 pp), whereas TRUE ends above it ($+5.245$ pp). PSEUDO’s worst-class point change is slightly worse than SYN’s, and no interval for that secondary contrast is reported here (see Table 8).

The PSEUDO comparison is not a pure, label-free support experiment. Self-targets can carry inherited task information and reinforcement effects. Likewise, TRUE minus PSEUDO includes subsequent changes in generators, states and selected actions. The contrasts do not identify independent causal mediators or warrant assigning a percentage of protection uniquely to labels. They establish the effects of the particular matched interventions.

What these investigations contribute. The selector’s score, the successor’s training information and the independent audit are different parts of the recursive system. Their operational roles can be separated even when no canonical sufficient statistic has been identified for the implementation. The failed criterion-specificity test, the exploratory harmful-correction findings and the completed target ablation constrain explanations of this substrate. They do not supply a universal collapse mechanism, show that original data are unnecessary, or validate F_β as a recursive controller.

9 Relationship to existing theories

The relevant theories differ in what must be preserved: a criterion posterior, an action comparison, a Bellman computation, or a controlled transition law. RCIT compares these requirements at

an interface and accounts for the information needed when that interface takes on an additional role.

Task-relative preservation. Classical statistical sufficiency and Blackwell comparison preserve inference or attainable decision value under reduced observations (Blackwell, 1953; Bahadur, 1954). Bruhacs (2026) explicitly formulates task families, factorisation through projections, minimal sufficient support and differences among prediction, explanation and update requirements. The query quotient in proposition 3.1 uses this preservation framework. Here the two query families specify current action comparisons and action-induced successor evaluation, with their residual connected to recursive and joint completion.

Fixed loss, complete regret and operating families. Bayes-sufficient representations retain the information needed for the optimum under a fixed loss. Loss shift can invalidate such a representation even when the joint distribution stays fixed (Sevetlidis, 2026a,b). TSC distinguishes this optimal-action endpoint from pointwise preservation of all feasible-action regrets and proves the corresponding whole-curve contraction guarantee. Hyeon et al. (2026) study action and Q-sufficiency, including optimal-policy and tie-breaking conditions. For matched consumers, the distinction relevant here is between absolute action values, relative action costs and the selected optimum.

Recurrent support and planner operations. Walsh (2026a) studies policies that alter subsequent inferential conditions and support representations that are adequate for one cycle but insufficient across recurrence. His single-cycle formulation uses action regret as distortion (Walsh, 2026b). The present analysis compares explicit decision and successor-evaluation query families and quantifies the supplement relative to the contract already supplied. Its finite completion results address exact preservation under a specified kernel, rather than regulation of a learned support resolution.

Wei et al. (2026) distinguish mechanism, candidate-response and decision sufficiency, and examine the richer information required while a planner searches for candidates. They identify common-offset-invariant relative-cost weights for MPPI updates. With fixed candidates and a matched criterion, absolute response costs determine the regret profile, which determines a unique optimum; planner-specific summaries can be coarser than the complete profile. Our same-task construction instead compares a current contract with the next decision geometry induced by action. The abstract of Counterfactual Quotient Models describes a related target of action-conditioned differences rather than unrestricted future prediction (Chen et al., 2026).

State abstraction and completion. The hierarchy of Li et al. (2006), stochastic bisimulation and MDP minimisation distinguish preservation of rewards, transitions, values and policies (Ferns et al., 2004; Givan et al., 2003). Dean–Givan minimisation covers any initial partition, including a task-derived seed (Dean and Givan, 1997). Complete-abstraction theory studies minimal refinement relative to semantic operations (Ranzato and Tapparo, 2007). Information-state theory separately requires performance sufficiency and predictive self-sufficiency, and provides exact and approximate control results and decentralized extensions (Subramanian et al., 2022). Our finite completion uses the coarsest-refinement construction and measures its information relative to the specified starting contract. It preserves the task-label process under quotient-implementable policies, as stated in Section 4.

Models for specified computations. Value equivalence preserves a declared family of Bellman computations rather than the full world model (Grimm et al., 2020). Rate–distortion formulations also select which planning distinctions to retain (Arumugam and Van Roy, 2022).

The conditional rate–distortion problem in Equation (15) takes the current decision contract as side information and the successor-evaluation label as its reconstruction target. This locates lossy supplementation within the established theory while distinguishing it from approximate dynamic closure.

Coordination and welfare. Coordination-capacity theory characterises communication needed to generate prescribed joint laws (Cuff et al., 2010). The contract formulation begins with the task answers that must remain recoverable; connecting the two requires an observation and communication model. The Gibbs action geometry in Section 7 is the finite rational-inattention optimisation of Armenter et al. (2024). Its Nash-product form describes a current decision tradeoff, while recursive adequacy is determined by the additional successor requirements.

10 Open problems in learning and coordination

The exact constructions identify what the declared interfaces must preserve. The next problems concern approximation, learning and access to the required information.

10.1 Approximate completion rather than lossy labels alone

For a finite interface Z refining D_0 , one possible closure error is

$$\epsilon_P(Z) = \max_{a, s, t: Z(s)=Z(t)} \|P_a(Z(S') \in \cdot \mid s) - P_a(Z(S') \in \cdot \mid t)\|_{\text{TV}}. \quad (50)$$

Here total variation is one half the ℓ_1 distance. The corresponding explicit design problem is

$$C_\mu^\epsilon(D_0) = \inf_{Z: D_0 \preceq Z, \epsilon_P(Z) \leq \epsilon} H_\mu(Z \mid D_0). \quad (51)$$

At $\epsilon = 0$ it reduces to the exact finite problem. At positive error, useful task or control guarantees require specifying loss sensitivity, horizon or discount and an admissible policy class. Existing approximate-information-state and bisimulation results provide the natural comparison (Subramanian et al., 2022; Ferns et al., 2004). Simply quantizing D_∞ need not solve Equation (51), because quantization can alter the new representation’s own transition consistency.

10.2 Learning the supplement and using finite evidence

A known kernel permits exact signatures. A learned kernel produces uncertain ones. A practical RCIT problem must distinguish a true task distinction from a distinction in a noisy estimate, and specify what evidence supports retaining or merging it. TRIC’s support budget makes this distinction explicit under its one-path assumptions. When those assumptions apply, requiring K distinguishable, locally supported classes also requires $Kq \leq M$. The mere fact that an entropy gap is positive does not show that the evidence budget can identify it.

A useful method would return a proposed supplement together with an evidence-dependent guarantee. Its validation problem is to determine which estimated distinctions are supported and which can be merged while retaining the required task answers.

10.3 Distributed supplements and shared memory

The quantity Γ_μ indicates a missing global distinction. To turn it into a network result, specify the agents’ observations, private or public memory, permitted messages, side information and coordination loss. A supplemental label can be mathematically small yet costly to acquire if no

single agent observes it. Equally, a receiver may reconstruct it from distributed observations without any sender transmitting that label literally.

Shared agent memory is a natural instance. A stored summary may be adequate for current local retrieval yet omit dependence required to evaluate joint actions or to update that summary faithfully. A rigorous study should compare matched task guarantees and memory or communication budgets, include an already-closed case, and distinguish the cost of computing the summary from the size of the exported interface. The five-state witness supplies one exact dependence failure against which such an architecture could be checked.

10.4 Action choice and future information requirements

A controller might prefer actions partly because their successors are easier to evaluate or coordinate. An explicit extension would attach a declared evaluation-cost term to the task objective. That cost must say whose information is available, which future queries matter and whether uncertainty in the estimated contract is included. Adding F_β , a closure entropy or its derivative to an objective without these choices does not automatically implement consequence-aware control. The failed surrogate-prediction tests are a reason to state the contract before selecting the scalar.

Performative prediction studies how deploying a predictor changes the distribution on which it is assessed (Perdomo et al., 2020). Identifying that action-induced distribution and determining which coarse interface preserves it are separate questions. A data-driven extension of the pullback construction needs transition or causal assumptions that identify the relevant interventional law.

10.5 Observable feedback and partial monitoring

For a finite partial-monitoring problem, let S stack the feedback signal matrices over actions and let p be an outcome distribution. The combined feedback law is Sp . Recovering an expected relative loss $(\ell_i - \ell_j)^\top p$ from that law for all p requires the appropriate loss difference to lie in the row span of S , under the usual signal-matrix conventions. This is the standard observability question (Bartók et al., 2014), read as sufficiency of a feedback-distribution interface for specified comparison queries. It concerns population feedback distributions, not recovery of a realised loss from a single observed symbol. This gives a concrete feedback-distribution instance of a task-relative comparison contract.

10.6 Continuous interfaces and changing welfare geometry

For continuous state, measurable quotients, invariant sigma-algebras, unbounded losses and finiteness of conditional information require additional hypotheses. Tracking a changing Gibbs utility region also raises questions about source weights, action-menu changes and continuation-relevant structure beyond one fixed-slope optimum. Finite rational-inattention theory supplies the starting geometry; connecting it to practical weighted-utility coordination requires a model of how those weights and utilities evolve.

11 Conclusion

A representation can remain exact for current action comparisons while omitting information needed to evaluate their successor circumstances. RCIT makes the two requirements explicit and asks whether the current contract determines the action-conditioned evaluation answers. The same-task construction shows that this implication can fail even when every current action regret is preserved and the same unique optimum is selected.

For finite deterministic interfaces, the missing information is $H(G_E^{\leftarrow} \mid G_D)$ under the declared source law. Classical controlled-Markov refinement supplies an exact completion for recursive use, with $H(D_\infty \mid D_0)$ measuring the supplement beyond the starting contract. Joining individually closed contracts can leave an additional $H(\text{Cl}_P(J) \mid J)$ because exact next marginals need not determine the joint next law. All three quantities apply the same contract-relative deficit to different evaluation targets.

These relationships place TRIC’s criterion contracts, TSC’s source reductions and RCIT’s recursive requirements within the common CIT framework. The exact constructions distinguish the requirements of current choice, recursive prediction and absolute-reward preservation. The Nash-welfare correspondence describes a current information–regret tradeoff, while the learning investigations examine particular routes for supplying training and evaluation information.

The resulting design sequence is to declare the current task, identify its sufficient interface, specify the successor questions, and establish what supplementation or completion they require. Learning, acquiring and communicating that supplement are the operational problems that follow. The distinction between these stages makes it possible to retain exactly the information required by a declared role while recognising when subsequent use demands more.

A Complete finite constructions and additional checks

A.1 Four-state chosen-action separation

The complete deterministic transition table supplementing Equation (11) is

Current state	action a	action b
x_1	y_1	x_1
x_2	y_2	x_2
y_1	y_1	y_1
y_2	y_2	y_2

The current regret partition over the full state set is $\{x_1, x_2, y_1\} \mid \{y_2\}$. The selected-action successor profile distinguishes x_2 from x_1 . The law uniform on the two current states gives one bit of residual evaluation information. The self-loop completion of unused transitions is only to specify a runnable finite model; the separation uses the two a transitions alone.

This example also distinguishes information held before acting from information acquired afterwards. If the next state is fully observed, the agent can read its next regret profile at that time. The theorem does not deny that possibility. It says the old current contract alone does not determine that profile in advance or preserve it without an additional observation.

A.2 Eight-state reduction

Table 1 supplies the entire reward and transition model. The initial regret partition is

$$\{p_1, p_2, p_3, p_4\} \mid \{q_1, q_2, q_3, q_4\}.$$

The stable four-class quotient has the transition table

Block	action a	action b
$P_1 = \{p_1, p_2\}$	P_1	Q_1
$P_2 = \{p_3, p_4\}$	Q_2	P_2
$Q_1 = \{q_1, q_2\}$	P_2	Q_1
$Q_2 = \{q_3, q_4\}$	Q_2	P_1

The reward-respecting partition is

$$\{p_1\} \mid \{p_2\} \mid \{p_3, p_4\} \mid \{q_1\} \mid \{q_2\} \mid \{q_3, q_4\}.$$

It is already stable. With the uniform raw law, the four quotient probabilities are $1/4$ each. The reward partition probabilities are four copies of $1/8$ and two copies of $1/4$, hence entropy 2.5 bits. Author-side numerical verification checked direct refinement, enumerated stable refinements of the starting partition and checked the projected paths for every deterministic stationary four-class policy through ten steps. Exact rational kernels establish the finite equivalence; the path enumeration is an additional implementation check, not the general proof.

A.3 Multi-stage closure

The five-state chain from Equation (29) has partitions and entropies shown in Table 4. The displayed entropy values are numerical evaluations of exact rational class probabilities.

Step	Partition	$H(D_k)$ (bits)	Increment (bits)
0	$\{s_0\} \mid \{s_1, s_2, s_3, s_4\}$	0.721928095	—
1	$\{s_0\} \mid \{s_1\} \mid \{s_2, s_3, s_4\}$	1.370950594	0.649022500
2	$\{s_0\} \mid \{s_1\} \mid \{s_2\} \mid \{s_3, s_4\}$	1.921928095	0.550977500
3	$\{s_0\} \mid \{s_1\} \mid \{s_2\} \mid \{s_3\} \mid \{s_4\}$	2.321928095	0.400000000

Table 4: Uniform-law information increments in a three-round closure. The total supplement is 1.6 bits.

A.4 Discounted shaping and its boundary conditions

For a fixed policy and initial state, the discounted shaping terms telescope:

$$\sum_{t=0}^T \gamma^t (\gamma \phi(S_{t+1}) - \phi(S_t)) = -\phi(S_0) + \gamma^{T+1} \phi(S_{T+1}).$$

Boundedness and $\gamma < 1$ make the last term vanish as $T \rightarrow \infty$. The return under every policy shifts by $-\phi(S_0)$. Conditioning on the first action gives the same shift for Q^π ; taking optimal values gives $G^{*,\phi} = G^*$. A finite-horizon or abruptly terminated process needs a terminal convention that cancels the residual potential. Without it, the stated invariance need not hold.

In the $2m$ self-loop family, $V^* = 1/(1-\gamma)$ at every state. At p_i , taking a is optimal and taking b loses exactly one unit before subsequent optimal behaviour. At q_i , the roles reverse. Thus G^* has exactly two rows, independently of m . Shaping subtracts $(1-\gamma)\phi(s)$ from both immediate rewards at s . Different i values separate all within-type absolute reward rows. Cross-type rows cannot coincide because the sign of the action difference is opposite. The reward-respecting partition is therefore discrete after shaping.

For a fixed μ , unchanged G^* and P imply unchanged closure information. If μ is generated by a fixed policy, its occupancy is also unchanged because the transition law and that policy are unchanged. A newly selected policy-dependent accounting law is a different comparison and must be specified rather than assumed invariant.

A.5 Joint marginal and joint-law checks

For the matrix in Equation (36), the probabilities of the first local block are

$$P(C_1(S') = 0 \mid s) = (1/2, 1/2, 3/4, 1/2, 3/4),$$

$$P(C_2(S') = 0 \mid s) = (1/2, 3/4, 1/2, 1/2, 3/4).$$

They are constant within the corresponding current local blocks. The next law of the joined label from states 0 and 3, ordered by blocks $\{0, 3\}, \{1\}, \{2\}, \{4\}$, is respectively

$$(1/6, 1/3, 1/3, 1/6), \quad (1/3, 1/6, 1/6, 1/3).$$

This is an exact difference in dependence left unresolved by the local marginals. The uniform law used for the entropy is an accounting distribution, not an assertion that it is stationary for the displayed chain.

For the independent product, summing its next product-block law over all but one coordinate recovers that coordinate's next-block law. This proves the reverse implication needed in proposition 6.2: equality of the full product-block laws forces equality of every marginal, rather than merely assuming that coordinatewise refinement is minimal. Author-side numerical verification checked the one- and two-copy constructions directly. The stated result for arbitrary k rests on the product proof, not numerical extrapolation.

B Finite information–welfare identities

All logarithms in this appendix are natural. These proofs make the finite correspondence self-contained. The exponentiated-action hull, weighted-log optimum and support bound are attributed to finite rational-inattention geometry (Armenter et al., 2024). The other steps use standard convex analysis and rate–distortion theory (Cover and Thomas, 2006).

B.1 Uniqueness, supporting actions and proportional fairness

At finite β , every component of every action vector is strictly positive. The compact hull $\mathcal{U}_\beta$ is therefore contained in the positive orthant. The function

$$G(u) = \sum_s p(s) \log u_s$$

is strictly concave because every $p(s) > 0$. It has a unique maximising utility vector u^* . The first-order optimality condition for maximisation over a convex set is

$$\nabla G(u^*)^\top (u - u^*) \leq 0 \quad (u \in \mathcal{U}_\beta).$$

Since $\nabla_s G(u^*) = p(s)/u_s^*$ and $\nabla G(u^*)^\top u^* = 1$, this is exactly Equation (47). It also gives, for every action,

$$\sum_s \frac{p(s)}{u_s^*} e^{-\beta d(s,a)} \leq 1. \quad (52)$$

If an action has positive mass in an optimal mixture, equality holds: the weighted average over that mixture equals one, so no positively weighted strict inequality is possible. The same condition follows from the mixture simplex KKT equations.

The optimum lies on the exposed face defined by $w^\top u = 1$, where $w_s = p(s)/u_s^* > 0$. This face has affine dimension at most $|\mathcal{S}| - 1$. Carathéodory's theorem within the face expresses u^* as a convex combination of at most $|\mathcal{S}|$ action vectors. The mixture realising it may not be unique. At $\beta = 0$, all action vectors are the all-ones vector, and a one-action mixture suffices.

B.2 Full-curve duality

The finite-channel feasible set is compact, expected regret is affine and mutual information is convex in the channel for fixed source law. The achievable rate–regret value $R_{\text{nat}}(\delta)$ is convex, nonincreasing and closed on its feasible domain. The fixed-slope value can therefore be written

$$F_\beta = \inf_{\delta' \geq 0} \{R_{\text{nat}}(\delta') + \beta \delta'\}.$$

The supporting-line representation gives

$$R_{\text{nat}}(\delta) = \sup_{\beta \geq 0} \{F_\beta - \beta\delta\}.$$

At a boundary point the supremum may require slopes tending to infinity rather than an attained finite slope. Substitution of Equation (44) gives Equation (48). This is the usual convex dual description of the curve, not a new rate theorem.

B.3 Slope and endpoint

F_β is an infimum of affine functions of β , hence concave. Nonnegative distortion makes it nondecreasing. At a differentiable slope, the envelope formula gives

$$F'_\beta = \mathbb{E}_{Q_\beta} d(S, A) = -\frac{d}{d\beta} \log \text{NSW}_\beta^*. \quad (53)$$

At nondifferentiable slopes, the appropriate statement uses one-sided slopes or the set of optimal distortions, rather than selecting an unjustified unique derivative. At $\beta = 0$, $F_0 = 0$ and $\text{NSW}_0^* = 1$. The right derivative is

$$F'_{0+} = \min_a \sum_s p(s) d(s, a),$$

the best state-independent expected regret.

To establish the hard-decision limit, choose a zero-regret channel and minimise its mutual information. It gives $F_\beta \leq R_{\text{nat}}(0)$. Hence the nondecreasing F_β converges to some $L \leq R_{\text{nat}}(0)$. For optimal channels Q_β ,

$$\beta \mathbb{E}_{Q_\beta} d \leq F_\beta \leq R_{\text{nat}}(0),$$

so expected regret tends to zero. Along a convergent subsequence of channels, continuity gives a limiting zero-regret channel Q_∞ . Since $F_\beta \geq I_{\text{nat}, Q_\beta}(S; A)$, it follows that $L \geq I_{\text{nat}, Q_\infty}(S; A) \geq R_{\text{nat}}(0)$. Thus

$$F_\beta \rightarrow R_{\text{nat}}(0), \quad \text{NSW}_\beta^* \rightarrow e^{-R_{\text{nat}}(0)}.$$

Under a unique zero-regret action at every positive-probability state, any zero-regret channel must set $A = a^*(S)$. Therefore $R_{\text{nat}}(0) = H_{\text{nat}}(a^*(S))$. With ties the zero-regret channel can exploit them, and the entropy of an arbitrarily fixed selected optimum need not equal $R_{\text{nat}}(0)$.

B.4 A closed-form numerical check

For a closed-form check, take two equiprobable states and mismatch regret $d(s, a) = \mathbf{1}\{s \neq a\}$. Symmetry gives the optimal common mixture $(1/2, 1/2)$ and

$$F_\beta = \log 2 - \log(1 + e^{-\beta}), \quad \text{NSW}_\beta^* = (1 + e^{-\beta})/2, \quad D_\beta^* = \frac{1}{1 + e^\beta}. \quad (54)$$

Author-side channel-grid and finite-difference computations checked these expressions numerically. These checks concern the standard finite example and are separate from the recursive-learning experiments.

C Experimental protocols and complete summary tables

C.1 Study design and evidential scope

The criterion-coupling experiment and target ablation use prospectively fixed designs, whereas the persistence analysis is explicitly post-hoc. The tables below retain the reported outcomes and inferential conventions for each investigation. Appendix D states the availability of the underlying empirical and finite-model verification artifacts.

C.2 Substrate, cycle, interventions and target provenance

Data and panels. Both 32-cycle campaigns use the canonical Fashion-MNIST release with 60,000 training and 10,000 test images in ten classes. Each image is flattened to 784 pixels, converted from unsigned bytes to single precision and divided by 255, then promoted to double precision for the receiver. The receiver consumes these pixels directly. The genuine transition source is a fixed 3,000-image training subset, with 300 images per class. The external steering panel contains 1,000 test images, 100 per class, drawn from the previously used 8,000-image assessment panel. The hidden panel contains its full 2,000-image test-set complement, 200 per class; it is not the earlier 8,000-image panel. The historical panel and steering subset use per-class NumPy seeds [2026091303, class] and [2026093001, class], respectively, with sorted source indices. Source IDs and exact pixel hashes were checked for separation of steering, hidden and training examples. ENDO instead draws 1,000 synthetic evaluation inputs separately for each role and cycle, labelled by that role’s raw parent and reused across its three candidate forecasts.

Receivers and training. Each unit contains six double-precision NumPy multilayer perceptrons, D1–D3 and A0–A2, with dimensions 784–256–10, one ReLU hidden layer and softmax cross-entropy training. The 20 units are the four seeds numbered 00–03 in each of five fixed lineages numbered 0–4. Their role-specific starting classifier and generator states are the frozen historical FP6 4R h1 RAW checkpoints, copied identically across conditions within each unit. The separate smoke unit is excluded. Corrections clone the current weights; each base replacement initializes a new classifier with Xavier-uniform weights and zero biases. Adam uses learning rate 0.001, moment coefficients 0.9 and 0.999, and numerical constant 10^{-8} , with fresh moments for every training call and a new row permutation each epoch. In the executed implementation, the configured weight-decay coefficient 0.0001 is added to the output-weight gradient only; hidden weights and biases receive no decay term. Each forecast and applied correction trains on 1,024 rows for three epochs in batches of 32 (96 updates). Each base replacement trains on 6,000 rows for 30 epochs in batches of 128 (1,410 updates), without early stopping. Candidate forecasts share their role’s probe stream; application uses a separate stream. Streams depend on unit, role, cycle and operation, with no condition coordinate. The frozen environment uses Python 3.12.3, NumPy 2.3.5 and SciPy 1.18.1, with single-thread numerical libraries.

Generator. Each role carries a class-conditional Gaussian generator in a fixed 32-dimensional PCA space. The inherited PCA mean and basis were obtained by centring and singular-value decomposition of the original 20,000-image training subset: 2,400 examples in each class except classes 3 and 9, which have 400 each. They are identical in all 120 starting generators and remain fixed during these campaigns. At each base transition, the generator refits class means, full sample covariances (divisor equal to class count minus one) and class frequencies on the selected training batch. A covariance ridge of 10^{-4} times the identity is added. A singleton class uses its observed mean and identity covariance before the ridge; an absent class has zero sampling probability. Sampling follows the fitted class frequencies and projects back to 784 pixels without clipping or whitening. PCA is used by the generator, not as a reduction of the receiver input.

Literal actions and selection. The fixed correction bank for each lineage contains 16,000 archived synthetic examples from generation 3. Each action concatenates eight ordered groups of 128 rows from that bank. Action 0 retains the original groups; actions 1 and 2 each replace one slot as listed below. Group and slot indices are zero-based. These complete 1,024-row actions remain fixed across all units in the lineage and all 32 cycles.

Lineage	Action 0 ordered group IDs	Action 1 slot / new group	Action 2 slot / new group
0	2, 10, 13, 33, 34, 35, 37, 39	4 / 3	3 / 20
1	2, 6, 15, 19, 21, 24, 33, 35	0 / 32	2 / 38
2	0, 3, 13, 15, 18, 19, 25, 28	3 / 22	1 / 27
3	10, 17, 18, 20, 26, 28, 30, 38	2 / 21	2 / 8
4	3, 5, 15, 21, 26, 32, 35, 39	2 / 37	6 / 18

For ENDO and EXTY, each candidate’s score is the equal mean of its ten class accuracies and A0–A2 roles; an absent reference class in an endogenous panel contributes zero. EXTY scores on the fixed external steering panel. EXTWRONG uses that same panel but weights classes 3 and 9 by one half each and all other classes by zero. All adaptive conditions train three forecast corrections for each of the six roles, but only A0–A2 scores enter selection. The selector receives three candidate scalars, transformed into a single regret row by subtracting each score from the maximum. Its native common-zero-action branch therefore chooses the maximum-score action with lowest-ID ties, and broadcasts it to all six roles. NOREVAL fixes action 0 and omits forecast training, while retaining the same applied-correction and replacement budgets.

Cycle and target placement. The inherited cycle begins with a raw current parent M_t . Three synthetic correction forecasts are evaluated using the condition’s declared panel and the fixed selection rule. The selected correction is applied, followed by the base transition to raw M_{t+1} . The prospectively frozen Phase-B protocol retains the native decision rule, including $\beta = 100$ and deterministic lowest-ID ties. This is an implementation setting, not a claim that the score is a sufficient recursive state.

The synthetic-only base transition uses 6,000 examples. The mixed condition uses 5,000 synthetic and 1,000 genuine examples. Every condition also retains the 1,024-example synthetic correction. The genuine pool for PSEUDO uses exactly the intended TRUE input IDs. Pseudo-targets are generated by the raw pre-decision parent before any forecast training and remain frozen. The 5,000 synthetic base labels continue to come from the selected corrected parent. The generator refit consumes the same combined transition batch, so target replacement affects subsequent generation as well as receiver training. Shared random streams pair innovations, not identical state-dependent generated values.

For each receiver role and cycle, the mixed-condition sampling stream first draws 5,000 indices from the newly generated 24,000-row synthetic pool and then 1,000 indices from the 3,000-row genuine-input source pool. Both draws are uniform without replacement within the cycle; both index sets are sorted, and synthetic rows precede genuine-input rows in the combined batch. The synthetic pool is distinct from the archived correction bank. The pseudo-targets are hard argmax class labels; no soft-label or distillation loss is introduced, and no additional candidate base-transition trainings are performed.

The protocol separates a feature-only genuine-input pool from withheld genuine targets in PSEUDO processes. Neither hidden-audit nor external-steering examples may enter its gradients. Genuine targets become available only to an offline assessor after all required trajectories have completed and been sealed. TRUE necessarily uses its authorised genuine targets. These are different provenance permissions, not a claim that all conditions are free of historical original-data influence.

The reuse rule requires the preselected unit 10_k00 to reproduce every raw and corrected checkpoint, generator, action, score, margin, seed and relevant path-independent provenance field through all 32 cycles in both SYN and MIX. Only then may all 20 historical controls be reused. Otherwise all three conditions must be rerun freshly and paired. The completed experiment used historical anchor mode. Availability of the replay receipts is specified in Appendix D.

C.3 Experimental units, endpoints and intervals

The original seed/unit is the experimental unit. There are four units in each of five fixed lineages, with six model roles nested within each unit. The primary capability C_t is the equal average of the ten fixed class scores and A0–A2 roles. The terminal preservation and normalized area are

$$\Delta C_{32} = C_{32} - C_0, \quad \bar{C} = \frac{C_0/2 + \sum_{t=1}^{31} C_t + C_{32}/2}{32}. \quad (55)$$

Whole-unit bootstrap draws resample four units with replacement inside each fixed lineage, weight lineages equally and preserve cross-condition, class and generation pairing. The resampling procedure uses 100,000 draws and seed 2026092401. The intervals condition on these lineages, source checkpoints and panels. Repeated actions, classes, receiver roles and evaluation examples are not additional independent replications.

The 97.5% intervals retain the prescribed two-primary-endpoint convention within each contrast. They do not establish simultaneous familywise control across all displayed comparisons and classification labels. Phase A is explicitly post-hoc and uses descriptive 95% intervals. No post-hoc equivalence criterion is attached to its uncertain differences.

C.4 Criterion-coupling outcomes

Condition	Terminal C	Preservation	Area $\bar{C}$	Worst-class change
SYN-ENDO	70.531	−3.837	71.415	−5.783
SYN-EXTY	70.699	−3.668	71.683	−5.892
MIX-ENDO	79.612	+5.245	79.802	+17.275
MIX-EXTY	79.740	+5.372	79.823	+15.508
SYN-EXTWRONG	70.937	−3.431	71.742	−3.692
SYN-NOREVAL	70.743	−3.624	71.721	−6.542

Table 5: All six criterion-coupling conditions. Capability is displayed on a percentage scale; changes are percentage points. Rounding of means can differ from rounded paired contrasts.

Contrast	Terminal difference (pp)	Area difference (pp)
SYN-EXTY − SYN-ENDO	0.168 [−0.337, 0.692]	0.268 [0.103, 0.440]
SYN-EXTY − SYN-EXTWRONG	−0.237 [−0.785, 0.237]	−0.060 [−0.213, 0.097]
MIX-ENDO − SYN-ENDO	9.082 [8.502, 9.687]	8.387 [8.244, 8.526]
MIX-EXTY − MIX-ENDO	0.128 [−0.162, 0.425]	0.022 [−0.030, 0.075]
Factorial interaction	−0.041 [−0.568, 0.507]	−0.246 [−0.420, −0.074]

Table 6: Primary 97.5% intervals. The interaction is the mixed-training evaluation effect minus the synthetic-training evaluation effect. Its area interval is below zero but lies inside the declared ± 1 pp equivalence margin. “Small at that resolution” is not the same claim as exactly zero interaction.

The misaligned channel improved its own class-3/9 area criterion over ENDO by 2.013 pp with primary interval [1.096, 2.959] pp. Thus it was an active alternative evaluation signal, not merely random noise. The strongest CC-A interpretation nevertheless failed because primary- Y specificity was not established. The completed classification remains CC-B/CC-D; the criterion-specificity gate for a minimal-bandwidth follow-on was not met.

Adaptive condition	Harmful / beneficial	Mean immediate effect (pp)	Descriptive 95% interval
SYN-ENDO	13 / 7	-0.356	$[-0.597, -0.112]$
SYN-EXTY	12 / 8	+0.011	$[-0.183, +0.212]$
SYN-EXTWRONG	14 / 6	-0.138	$[-0.409, +0.135]$

Table 7: First adaptive-minus-persistent applied-child effects. There are 20 paired units per comparison, no tied first-switch outcomes in these data, and no claim that all 60 forks are independent replications.

C.5 First-switch audit

The first-switch audit checked common preceding action histories and six-role parent identities. Twenty-six generation-one forks use direct rehashes of original checkpoint sources matched to sealed copies; the other 34 use the original audit/pruning chain. Applied children share application seeds and menus. Forecast children used different probe seeds and lack hidden scores. Their hidden scores cannot be recovered from the retained artifacts. Immediate results therefore concern the applied correction, and later horizons concern diverging whole policies.

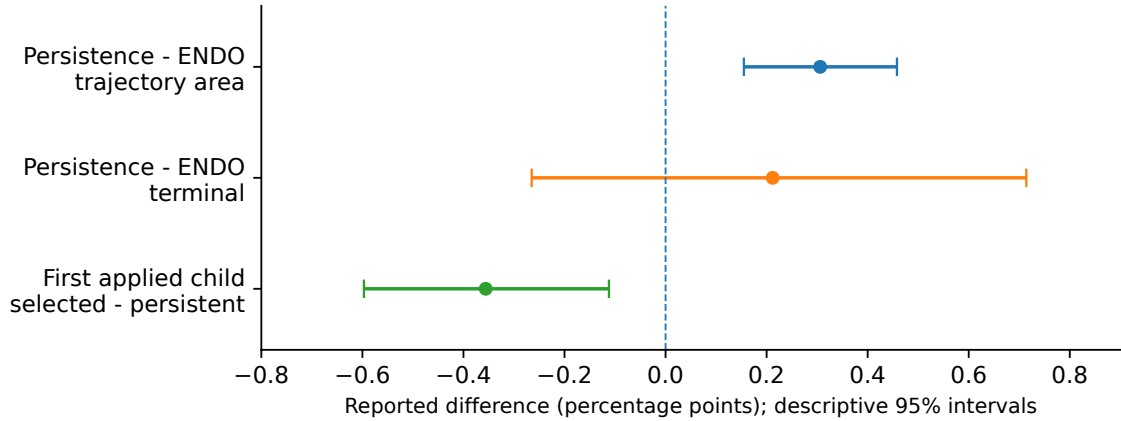


Figure 5: Selected Phase-A contrasts with descriptive 95% intervals. The two persistence contrasts are NOREVAL minus ENDO; the fork contrast is selected adaptive child minus persistent child. Directions differ by design and are labelled explicitly. The figure displays the reported summary estimates; it does not recompute the bootstrap analysis.

C.6 Target ablation outcomes and classification

Condition	Terminal C	Preservation	Area $\bar{C}$	Worst-class change
TRUE-ENDO	79.612	+5.245	79.802	+17.275
REALX-PSEUDOY-ENDO	71.387	-2.981	72.454	-6.058
SYN-ENDO	70.531	-3.837	71.415	-5.783

Table 8: All target-ablation conditions. Better mean preservation in PSEUDO does not establish better worst-class preservation. An interval for the corresponding worst-class contrast is not reported here.

The fixed interpretation rule defines benefit through a primary lower bound above zero, positive means in at least four of five lineages and the other endpoint’s lower bound above -1 pp. Substantial benefit uses a lower bound above +1 pp. Equivalence requires both primary

intervals strictly inside $(-1, +1)$ pp. Classification checks harm, label-dominance with support equivalence, support-dominance with label equivalence, both benefits, then unresolved, in that order. B-C requires the two positive contrasts on a common endpoint and no earlier qualifying class. These are prospectively frozen operational categories. They do not assert a unique causal decomposition.

The TRUE minus PSEUDO contrast is substantially positive. PSEUDO minus SYN is detectably positive but neither its terminal nor area interval has a lower bound above one percentage point; both endpoints also fail the required equivalence enclosure. B-C therefore follows the stated rule. It does not imply equal contributions.

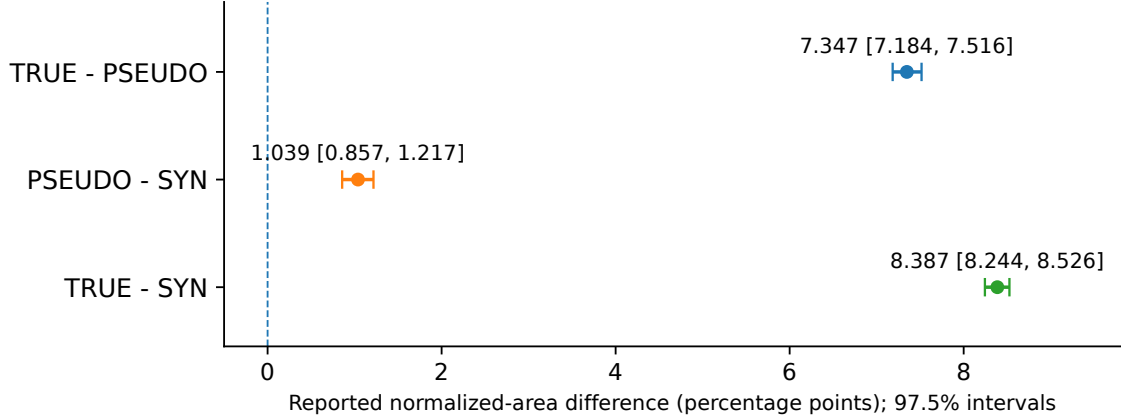


Figure 6: Phase-B normalized-area contrasts with primary 97.5% intervals. The larger genuine-target increment and the smaller positive genuine-input/self-target increment are both retained. This figure displays the paired summary estimates in Table 3, rather than reconstructed trajectories.

Pseudo-label diagnostics were assessed offline after all treatment trajectories were complete and sealed. Across 20 whole units, the Spearman correlation between mean pseudo-label accuracy and the mean TRUE minus PSEUDO child gap is -0.9263157895 . This is a secondary association. Both terms depend on the evolving model, so the correlation is not an isolated mediation estimate or a validated predictive controller. Per-generation, per-role and per-action measurements do not supply independent extra replications.

C.7 Earlier structural summaries

The earlier action-conditioned analysis covers 160 parents, 480 matched three-action children and 20 original units in five lineages. Its inherited balanced 8,000-example panel is reused. The subsequent distinction-level analysis retains complete feasible-action comparisons and separately records point estimates, uncertainty from the reused panel and subsequent capability. Its 5/5 exposure and approximate-current-value witness counts do not make the estimated classes canonical population quotients.

In particular, J is the constant deterministic action-channel objective, not the optimum F over stochastic channels. The equal-count witness $J = 133/160$ concerns the estimated child geometry in that analysis. Both child point partitions are discrete, so the observation is not a proof of a new partition split. Conditional panel intervals do not estimate repeated-training uncertainty. Negative frozen continuation gates for induced F , unresolved distinctions and temporal derivatives remain negative; they do not validate a further controller.

D Verification and data availability

The analytical proofs and complete finite-model specifications are included in this manuscript. Author-side numerical verification checked rational transitions for the four-state witness, eight-state refinement, multi-depth chain, five-state joint example and its two-copy product. Exhaustive enumeration of partitions of the relevant small state sets provided a separate implementation check of the coarsest-stable-refinement result. The proofs cover general finite systems and arbitrary independent product counts. Channel-grid and finite-difference calculations also checked the closed-form binary information-welfare example.

Aggregate empirical results, the reported experimental design and the analysis conventions are given in Appendix C. The empirical figures display the recorded estimates and intervals; their preparation and the finite-model checks did not rerun empirical training or bootstrap analysis.

The verification and experimental code, checkpoints, replay receipts and unit-level empirical outputs are not publicly released with this preprint version. No end-to-end empirical reproduction package accompanies this version.

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